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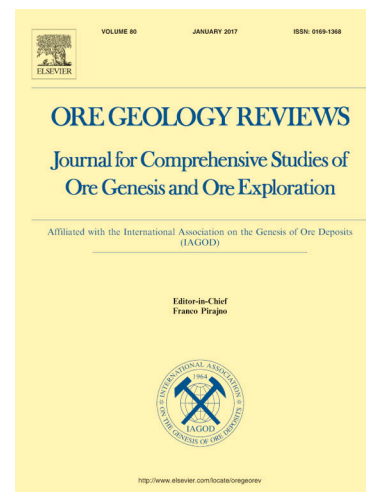
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Skarn geochronology and genesis of the Wanlongshan Zn-Sn polymetallic deposit in southeast Yunnan: Constraints from garnet and sphalerite composition, and in-situ S isotopes

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Abstract: The Wanlongshan Zn-Sn polymetallic deposit represents a significant deposit discovered recently in the world-class Dulong super-large Zn-Sn-In polymetallic ore field, situated in proximate to the Tongjie-Manjiazhai ore deposit. The stratiform to lenticular ore bodies are hosted within siliceous carbonate rocks of the Cambrian Tianpeng Formation's second member and exhibit structural control by interlayer fault zones. Despite its importance, the precise timing of magmatic-hydrothermal activity and the associated ore-forming processes remain incompletely constrained, generating considerable debate regarding its genesis. To address these uncertainties, this investigation employs an

integrated analytical approach combining garnet U-Pb geochronology, trace element geochemistry of garnet and sphalerite, and in situ sulfur isotope systematics. Geochemical analyses reveal that the garnets are predominantly grossularitic in composition, characterized by distinctive LREE depletion and HREE enrichment patterns that likely reflect the interplay between crystal-chemical constraints and adsorption mechanisms. The subdued europium anomalies suggest mineral formation under conditions of low oxygen fugacity within a weakly oxidizing to reducing environment that was neutral to mildly acidic, where hydrothermal metasomatism proceeded primarily via diffusional processes under restricted water/rock ratios. Geochronological constraints from garnet U-Pb dating yield an age of 88.1 ± 3.3 Ma (MSWD=0.98, $n=17$), which correlates remarkably well with molybdenite Re-Os ages from the mining district, ore formation ages at Manjiazhai, and the emplacement chronology of the Laojunshan granite. This temporal coincidence strongly indicates synchronicity among skarnization processes, Zn-Sn mineralization, and granitic magmatism. Trace element distributions in sphalerite reveal substantial enrichment in Fe, Mn, and In, with concomitant depletion in Ga, Ge, and Sn. Elemental ratios including Fe/Zn, Zn/Cd, and Ga/In in sphalerite indicate intermediate to high-temperature mineralization conditions. Multiple substitution mechanisms have been identified: Fe, Mn, and Cd incorporate through simple isomorphic replacement of Zn; Pb occurs predominantly as microinclusions; Cu enters as chalcopyrite solid solution; silver likely substitutes through coupled mechanisms involving either $\text{Ag}^+ + \text{Ga}^{3+} \rightarrow 2\text{Zn}^{2+}$ or $2\text{Ag}^+ + \text{Ge}^{4+} \rightarrow 3\text{Zn}^{2+}$; gallium exhibits minimal incorporation; while indium predominantly substitutes via $\text{Cu}^+ + \text{In}^{3+} \leftrightarrow 2\text{Zn}^{2+}$. The trace element signature of sphalerite differs markedly from patterns characteristic of Mississippi Valley-Type (MVT), Volcanic-Hosted Massive Sulfide (VHMS), and Sedimentary Exhalative (Sedex) Pb-Zn deposits, instead exhibiting geochemical affinities with established skarn deposits worldwide. In situ sulfur isotope analyses yield $\delta^{34}\text{S}$ values ranging from 0 to 3.8‰ (mean 2.1‰, $n=38$), consistent with predominant derivation from magmatic reservoirs, confirming that sulfur in the ore-forming fluids originated primarily from magmatic-hydrothermal sources. Collectively, these multiple lines of evidence—chronological, mineralogical, geochemical, and isotopic—provide compelling support for the classification of the Wanlongshan Zn-Sn polymetallic deposit as a classic skarn-type mineralization system.

Keywords: U-Pb dating of garnet; Trace elements in garnet and sphalerite; S isotope; Skarn deposit; Wanlongshan Zn-Sn polymetallic deposit.

1. Introduction

Stratiform skarn deposits are significant sources of resources in China, including Cu, Pb, Zn, Fe, Au, Mo, W, and Sn (Mao et al., 2011; Zhao et al., 2017; Chen et al., 2022). Representative deposits include the Dongguashan Cu-Ag deposit in Anhui, the Wushan Cu deposit in Jiangxi, and the Laojunshan Zn-Sn-In deposit in southeastern Yunnan. Due to the stratiform to lenticular occurrence of these skarn bodies, their syngenetic mineralization characteristics, and their spatial-temporal relationship with intrusive bodies, the genesis of these deposits has been widely debated. Proposed theories include magmatic-hydrothermal (Liu, 2021; Xiao et al., 2021), sedimentary exhalative (Zhou et al., 1998; Wang et al., 2007), and superimposed models (Liu et al., 1998; Xie et al., 2020). Despite the economic importance of these deposits, the lack of consensus on their formation mechanisms has limited the development of effective exploration models and hindered our understanding of regional metallogenic processes.

The Wanlongshan Zn-Sn polymetallic deposit, a critical component of the Dulong super-large Zn-Sn-In polymetallic ore field, is spatially adjacent to the Tongjie-Manjiazhai deposit. Cumulative to 2017, systematic exploration involving over 60,000 meters of drilling delineated 27 million tons of industrial ore, including 1.3 million tons of Zn and 40,000 tons of Sn, classifying it as a large deposit with significant economic value. Preliminary studies suggest a close relationship between the ore-forming process, granitic porphyry, and skarnization (Ai et al., 2018). The ore bodies primarily occur in a stratiform

to lenticular pattern within marble and schist layers controlled by interlayer fault zones, rather than directly at the contact zones between rock massifs and surrounding rocks. This pattern is similar to the Tongjie-Manjiazhai deposit. Despite extensive research on the Tongjie-Manjiazhai deposit, the genesis remains debated, with theories including magmatic-hydrothermal origin (Song, 1989; Liu et al., 2007; Ye et al., 2016; Liu, 2021), sedimentary exhalative (Zhou et al., 1998; Jia et al., 2014), and sedimentary-metamorphic-hydrothermal modifications (Liu et al., 1998). Determining the correct genetic model for the Wanlongshan deposit has significant implications for regional exploration strategies and for understanding similar deposits worldwide.

In recent years, advancements in LA-ICP-MS micro-analysis techniques have positioned hydrothermal minerals, particularly garnet and sulfides, as optimal samples for elucidating ore deposit origins (Fan, 2018; Zhou et al., 2019). These techniques facilitate precise in-situ elemental analysis and U-Pb dating of garnet (Seman et al., 2017; Wafforn et al., 2018; Tang et al., 2021; Liu et al., 2021; Sun et al., 2023), yielding high-precision trace element data for sulfides. This data accurately reflects trace element enrichment patterns. Combined with temporal resolution profiling, these methods reveal various states of trace elements within minerals (Cook et al., 2009; Wei et al., 2019; Hu et al., 2021). Furthermore, in-situ S isotope analysis of sulfides can determine the source of sulfur in ore deposits, the mechanisms of sulfur formation, and trace ore-forming processes, providing insights into deposit genesis (Yuan et al., 2015; Bao et al., 2017). To address the knowledge gap regarding the genesis of the Wanlongshan Zn-Sn polymetallic deposit, this study applies advanced LA-ICP-MS analysis with three main objectives: 1) to conduct U-Pb dating of garnet in skarn ore from the Wanlongshan Zn-Sn polymetallic deposit, determining the precise age of skarn formation; 2) to perform trace element analysis on garnet and sphalerite, discussing their formation processes, mechanisms of trace element enrichment, and physical and chemical conditions, thereby enhancing the understanding of the deposit's genesis; 3) to analyze S isotopes of sulfides within the deposit to clarify the source of mineralizing materials. By integrating these multiple lines of evidence, this study aims to resolve the controversy surrounding the deposit's formation and contribute to more effective exploration models. Our findings suggest that the Wanlongshan deposit is a typical skarn deposit with a clear magmatic-hydrothermal origin, as will be demonstrated through the geochemical and geochronological evidence presented in the following sections.

2. Regional Geological Context

The Wanlongshan Zn-Sn polymetallic deposit, located in southeastern Yunnan Province, lies on the southwestern edge of the Cathaysia Block, at the intersection of the Yangtze, Indosinian, and North Vietnam blocks (Fig. 1). It is adjacent to the Tethyan-Himalayan orogen to the west and the Youjiang Rift Basin to the north, forming part of the North Vietnam Block marginal trough zone (Roger et al., 2000; Maluski et al., 2001). The Cambrian strata are exposed in the area, mainly composed of schist, gneiss, argillaceous clastic rocks, and carbonate rocks, appearing in the form of interbed. The Maguan-Dulong and Wenshan-Malipo faults are the primary structures controlling mineralization in the Laojunshan ore cluster, where the Wanlongshan deposit is situated within the triangular zone defined by these two faults. The area has experienced significant magmatic activity, with prominent outcrops of the Caledonian Nanwenhe gneissic granite and the Yanshanian Late Laojunshan granite. The Laojunshan granite, covering approximately 134 km², extends about 14 km north-south and 9 km east-west and dips southward into the study region. It can be divided into three magmatic phases (Fig. 2a): **1. Initial Phase:** Medium to coarse-grained two-mica monzogranite, aged 118–110 Ma, occurs as batholiths distributed in the peripheral zones of composite plutons. **2. Subsequent Phase:** Medium to fine-grained two-mica granite, aged 98–90 Ma, intrudes as stocks into the primary phase. **3. Terminal Phase:** Porphyritic granites and quartz porphyries, aged 88–80 Ma, mainly appear as dikes intruding the earlier phases and Cambrian cover strata (Liu et al., 2007; Lan et al., 2016; Liu, 2021; Tian and Yang, 2023). The area is

rich in mineral resources, including the Dulong super-large Zn-Sn polymetallic mining district (Tongjie, Manjiazhai, Wanlongshan, Nandangchang), the Nanyangtian large W deposit, and the Xinzhai large Sn deposit, making it one of China's key Sn-In polymetallic resource bases. Previous studies suggest that W-Sn mineralization in the area is closely associated with the Laojunshan granite (Yang et al., 2019).

3. Ore deposit geology

3.1. Geologic Background

The Wanlongshan Zn-Sn polymetallic deposit exhibits a relatively straightforward geological setting, dominated by middle Cambrian strata from the second, third, and fourth phases of the Tianpeng Formation (Fig. 2b).

Among these, the second phase represents the most significant mineral-bearing horizon and comprises two distinct sub-stages. The first sub-stage (C_2t^{2-1}) consists primarily of schist with subordinate marble and skarn, occasionally interlayered with minor epidote-bearing plagioclase metamorphic granular rock, altered biotite plagioclase metamorphic granular rock, and metamorphic fine-grained quartz sandstone. Despite its complex lithology, skarn serves as the principal host rock for Sn and Zn mineralization in this deposit. The second sub-stage (C_2t^{2-2}) is dominated by marble with intercalated schist and skarn, but contains fewer mineralized Zn-Sn ore bodies compared to the first sub-stage.

The mining district occupies the western flank of the Laojunshan composite anticline, characterized by a predominantly west-dipping monocline with secondary faults and fold structures. Structural control is primarily fault-related, with the F_0 and F_1 faults being the most prominent (Fig. 2b). The F_1 fault, a reverse fault, extends northward to Tongjie and southward through the mining district, with most ore bodies situated along its footwall. This fault intersects with the F_0 fault to the north and continues southward before terminating at Nandangchang. It represents the main structural feature controlling ore emplacement and fracturing in the western segment of the Dulong Zn-Sn ore field.

The mining district lies near the southwestern margin of the Laojunshan granite. Surface and shallow intrusive outcrops are sparse, limited to isolated exposures of the third phase of Cretaceous Laojunshan granite (γ_5^3). Geophysical evidence suggests the presence of a blind granite body at depths exceeding 1200 meters. The concealed pluton shows geochemical characteristics similar to the second phase of the Laojunshan granite, with elevated concentrations of ore-forming elements, suggesting it served as the primary source of metallogenic material and energy.

3.2. Mineralization and hydrothermal alteration

Engineering work has identified 121 industrial ore bodies, with WL3, WL4, WL5, WL10, and W20 being the most significant (Fig. 3). These ore bodies do not outcrop at the surface, range from 40 to 960 meters in length, and measure between 0.68 to 24.23 meters in thickness. Zinc grades vary from 2.65% to 21.66% (averaging 4.63%), while tin grades range from 0.45% to 2.41% (averaging 0.58%). The ore bodies trend north-south in plan view and display a stratified, imbricated structure in profile, dipping southwestward at angles between 6° and 18° . The WL5 ore body, a major focus of this investigation, is stratified within the upper section of the first sub-stage of the second phase of the Tianpeng Formation (C_2t^{2-1}). It extends 720 meters in length and dips westward at depths between 92 and 374 meters, with an average dip of 26° .

Skarn constitutes the predominant ore type, featuring a complex mineral assemblage. The ores display massive, disseminated, and banded textures. Major metallic minerals include marmatite, accompanied by cassiterite, chalcopyrite, and pyrite. Gangue minerals consist mainly of diopside, chlorite, and garnet, with minor actinolite, epidote, and tremolite, typically exhibiting subhedral or euhedral textures. Hydrothermal alteration surrounding the ore bodies is pronounced, predominantly

represented by skarnization, with secondary fluoritization, silicification, and carbonatation. The emplacement of all industrial Zn-Sn ore bodies within late-stage skarns indicates a strong genetic relations.

Based on geological observations from gallery exposures and drill cores, combined with microscopic examination of mineral paragenesis, the mineralization process at Wanlongshan can be divided into three main stages comprising six sub-stages (Fig. 4): (1) Skarn Stage: a) Early Skarn Sub-stage: Characterized by anhydrous minerals including garnet, diopside, and minor wollastonite. The garnet typically occurs at marble contacts, displaying light brown coloration, euhedral granular texture, and positive relief without cleavage. Granular diameters reach up to 1.0 cm, lack zoning, appear isotropic under cross-polarized light with first-order gray interference colors, and show minimal mineralization. b) Late Skarn Sub-stage: Early skarn minerals are replaced by hydrous silicates such as sericite, tremolite, epidote, actinolite, and chlorite, forming reaction rims, with concomitant precipitation of fine-grained magnetite. c) Oxide Sub-stage: Marked by minor cassiterite, arsenopyrite, scheelite, pyrite, and magnetite, with associated calcite and subordinate quartz. (2) Hydrothermal Mineralization Stage: a) Quartz-Sulfide Sub-stage: Quartz-fluorite veins cut through early skarn, with sericitic alteration halos developed along their margins, and contain sulfides including pyrite, sphalerite, magnetite, chalcopyrite, and galena. b) Quartz-Carbonate Sub-stage: Involves hydrothermal fluids infiltration along structural fractures and fault zones, altering pre-existing silicate minerals and precipitating gangue minerals (quartz, calcite) and metallic sulfides (pyrite, sphalerite, galena). (3) Supergene Stage: Dominated by oxidation processes resulting in the formation of secondary minerals such as hemimorphite, chalcocite, and covellite.

4. Samples and analytical methods

4.1. Samples

For this investigation, samples were obtained from the WL5 ore body within the ZK2014 drilling of the Wanlongshan mining district, which extends northward into the Manjiazhai mining district (No.13 ore body). The ore mineralogy predominantly comprises metallic minerals (e.g., marmatite, chalcopyrite, bornite, pyrite, and cassiterite), and gangue minerals (e.g., garnet, diopside, and chlorite).

Following comprehensive microscopic examination, hand specimens designated as 2[#], 15[#], and 52[#] were processed into thin sections appropriate for analytical procedures. Through detailed microscopic observation, garnets exhibiting minimal mineral inclusions and well-developed crystal morphology were selected for in-situ U-Pb geochronological analysis. Concurrently, garnets and sphalerites characterized by pristine surfaces and minimal fractures were identified for in situ trace element compositional analysis via Laser Ablation Inductively Coupled Plasma Mass Spectrometry (LA-ICP-MS). Furthermore, Laser Ablation Multi-Collector Inductively Coupled Plasma Mass Spectrometry (LA-MC-ICP-MS) was employed for sulfur isotope analysis at locations proximal to the trace element measurement points on sphalerite and pyrite. (Fig. 5). Considering the relatively homogeneous paragenetic sequence of garnet and sphalerite throughout the Wanlongshan mining district, and as the evolution of ore-forming fluids falls beyond the scope of this investigation, a detailed paragenetic classification of these minerals was not undertaken.

4.2. Analytical methods

We performed three key analytical procedures at the State Key Laboratory of Ore Deposit Geochemistry (Chinese Academy of Sciences): (1) LA-ICP-MS U-Pb isotope dating, (2) in-situ elemental analysis of garnet and sphalerite, and (3) in-situ sulfur isotope analysis. The instrumentation for garnet U-Pb isotope dating and in-situ elemental analyses comprised a GeoLasPro 193 nm excimer laser ablation system coupled with a Thermo Element XR high-resolution ICP-MS. Analytical parameters included a 32 μm laser spot diameter, 5 J/cm² energy density, 5 Hz repetition rate, with He as the carrier gas and Ar as the auxiliary gas. Each analysis cycle consisted of 20s background measurement followed by a 45s laser ablation. Prior to sample dating, calibration was performed twice in the sequence of NIST SRM 612, 91500, Willsboro, and

QC04. During sample analysis, zircon standard 91500 (1062 Ma) functioned as the primary reference material, with garnet standard QC04 integrated periodically for quality control, additional standard measurements were conducted after every 10-15 analysis points. The analytical protocol followed procedures described by Tang et al. (2020, 2021).

Data reduction was executed using ICPMSDataCal software (Liu et al., 2010), while concordia diagrams for U-Pb ages were plotted using Isoplot 4.15, with reported uncertainties on isotope ratios and ages representing 1σ . For garnet trace element analysis, international standards GSE-1G, SRM 610, BIR-1G, BCG-2G, and BHVO-2G were utilized in a no internal standard–multiple external standard calibration method. Detection limits for most trace elements reached 0.001×10^{-6} , with an analytical precision exceeding 10%. Comprehensive analytical protocols and data processing methodologies are detailed in Liu et al. (2008).

The trace elemental analysis of sphalerite was executed using a Compex-Pro 193 nm laser ablation system (Coherent, Germany) coupled with an Agilent 7700x ICP-MS.

Helium served as the carrier gas during the laser ablation process, transporting the aerosol via a T-connector where it was mixed with argon gas during ablation, transporting the aerosol via a T-connector where it merged with argon prior to ICP-MS introduction. The laser spot diameter was primarily 25 μm , with an energy density of 17.3 J/cm² at a pulse frequency primarily set at 5 Hz, with ablation lasting for 60 s. Quantification of chalcophile and siderophile element concentrations employed the STDGL3 standard (Danyushevsky et al., 2011), while GSD-1G served as the normalization standard for lithophile element count data transformation. The MASS-1 sulfide reference material was analyzed as an unknown sample to validate analytical precision (Wilson et al., 2002). Data reduction was performed offline using LADR software (developed at CODES), achieving.

The experimental apparatus for in situ sulfur isotope analysis of sulfides comprised a RESOLUTION S-155 laser ablation system (Reso Company). The system utilized an ArF excimer laser generating a 193 nm deep ultraviolet beam, which was homogenized and focused onto the sulfide surface. Analytical parameters included a 40 μm laser spot diameter with an ablation frequency of 5 Hz for 40 s duration. A gas mixture of high-purity helium, argon, and nitrogen functioned as the carrier system for introduction into the mass spectrometer. To mitigate matrix effects, pyrite, magnetite, and nickeline samples were calibrated against pyrite reference material PPP-1.

Chalcopyrite samples were calibrated using the national standard material GBW07268 (in powdered pellet form), while silver sulfide samples were calibrated using the international silver sulfide standard material IAEA-S-1 (also in powdered pellet form). The recommended $\delta^{34}\text{S}_{\text{V-CDT}}$ values for the aforementioned samples are detailed in Fu et al. (2016). Throughout the testing procedure, in-house reference materials including magnetite SP-Po-01 ($\delta^{34}\text{S}_{\text{V-CDT}} = 1.4 \pm 0.4$), chalcopyrite SP-CP-01 ($\delta^{34}\text{S}_{\text{V-CDT}} = 5.5 \pm 0.3$), and international silver sulfide standard materials IAEA-S-2 ($\delta^{34}\text{S}_{\text{V-CDT}} = 22.58 \pm 0.39$) and IAEA-S-3 ($\delta^{34}\text{S}_{\text{V-CDT}} = -32.18 \pm 0.45$) were repeatedly analyzed to serve as quality control samples, thereby validating the precision of the experimental approach. Detailed instrumental parameters and analytical protocols are described in Fu et al. (2016).

5. Analytical results

5.1. U-Pb Chronology of Garnets

The principal isotopic signals (^{238}U , ^{206}Pb , ^{207}Pb , ^{202}Hg , ^{204}Pb , and ^{232}Th) demonstrated consistent stability throughout data acquisition (Fig. 6a). From the 21 U-Pb isotopic analyses performed on specimen No.15[#], 17 valid data points were retained after excluding unstable measurements (Table 1). The lead content $w(\text{Pb})$ ranged from 0.03×10^{-6} to 1.52×10^{-6} , averaging 0.31×10^{-6} . The thorium content $w(\text{Th})$ varied from 0.03×10^{-6} to 0.45×10^{-6} , with an average of 0.12×10^{-6} . The uranium content $w(\text{U})$ ranged from 0.12×10^{-6} to 2.79×10^{-6} , averaging 1.03×10^{-6} . The $^{207}\text{Pb}/^{206}\text{Pb}$ ratios ranged from 0.2068 to 1.2209, the $^{207}\text{Pb}/^{235}\text{U}$ ratios ranged from 0.3460 to

350.7719, and the $^{206}\text{Pb}/^{238}\text{U}$ ratios ranged from 0.0175 to 2.9360. The Tera-Wasserburg concordia diagram method was used for common lead correction (Tang et al., 2020, 2021), and Isoplot software was used for diagram construction. The lower intercept age of garnet in the Tera-Wasserburg (T-W) diagram was determined as 88.1 ± 3.3 Ma (MSWD = 0.98, $n = 17$; Fig. 6b).

Table 1

LA-sf-ICP-MS U-Pb dating results of garnet from the Wanlongshan Zn-Sn polymetallic deposit

Sample No.	Content ($\times 10^{-6}$)			Th/U	Isotope ratio					
	Pb	Th	U		$^{207}\text{Pb}/^{206}\text{Pb}$	1σ	$^{207}\text{Pb}/^{235}\text{U}$	1σ	$^{206}\text{Pb}/^{238}\text{U}$	1σ
15#-1	0.09	0.45	1.88	0.24	0.3884	0.0553	1.1689	0.2241	0.0225	0.0024
15#-2	0.11	0.44	2.79	0.16	0.2725	0.0472	0.7176	0.1626	0.0184	0.0013
15#-4	1.52	0.07	0.12	0.60	0.8654	0.0232	350.7719	31.2626	2.9360	0.2523
15#-6	1.00	0.08	0.24	0.34	0.8877	0.0401	118.2664	12.1896	0.9863	0.1000
15#-7	0.18	0.05	0.31	0.16	0.9016	0.0922	16.5026	2.9340	0.1421	0.0231
15#-8	0.03	0.06	0.75	0.08	0.3102	0.0844	0.6729	0.1550	0.0200	0.0021
15#-9	0.18	0.10	0.76	0.13	0.7206	0.0733	5.8864	0.4132	0.0614	0.0043
15#-10	0.11	0.06	0.81	0.08	0.6709	0.1300	2.9446	0.3336	0.0367	0.0036
15#-12	0.05	0.05	0.65	0.08	0.5953	0.1384	1.5398	0.2780	0.0240	0.0026
15#-13	0.03	0.06	0.51	0.12	0.2068	0.1301	0.3460	0.1462	0.0175	0.0024
15#-14	0.03	0.03	0.77	0.04	0.3301	0.1351	0.4831	0.1122	0.0175	0.0020
15#-15	0.65	0.08	0.93	0.09	1.2209	0.9660	18.0025	7.1151	0.1566	0.0588
15#-16	0.04	0.05	0.65	0.08	0.5835	0.5151	1.3985	0.4434	0.0214	0.0031

15 [#] -17	0.06	0.09	1.43	0.07	0.3588	0.1234	0.7385	0.1099	0.0186	0.0018
15 [#] -19	0.06	0.06	1.01	0.06	0.3353	0.0719	0.9242	0.1558	0.0217	0.0023
15 [#] -20	1.10	0.09	1.49	0.06	0.5421	0.0618	19.7065	5.2192	0.1834	0.0445
15 [#] -21	0.11	0.19	2.37	0.08	0.2943	0.0513	0.7483	0.1270	0.0188	0.0016

5.2. Trace element composition of garnet

The garnet in the Wanlongshan Zn-Sn polymetallic deposit exhibits trace element concentrations with Rb ranging from 0.001×10^{-6} to 0.76×10^{-6} , averaging 0.12×10^{-6} , Ba from 0.001×10^{-6} to 0.23×10^{-6} , averaging 0.05×10^{-6} , and Sr from 0.04×10^{-6} to 0.64×10^{-6} , averaging 0.11×10^{-6} (Table 2). Generally, the garnet shows significant depletion in LILEs (e.g., Rb, Ba, and Sr) and enrichment in HFSEs (e.g., Ta, Th, and U) (Table 2, Fig. 7). Elevated concentrations of Zn and Sn are also observed, with Sn ranging from 946×10^{-6} to 4788×10^{-6} , averaging 3022×10^{-6} , whereas Mo and Pb are comparatively lower.

The chondrite-normalized rare earth element (REE) distribution patterns demonstrate enrichment in heavy REEs (HREEs) and depletion in light REEs (LREEs). This pattern, where Σ REE (total REE content) ranges from 2.14×10^{-6} to 11.8×10^{-6} (averaging 5.06×10^{-6}), is characteristic of certain hydrothermal environments. The (La/Yb)_N ratio varies from 0.01 to 0.38, and the LREE/HREE ratio fluctuates between 0.42 and 5.35, with a minor Eu anomaly ($\delta\text{Eu}=0.29$ to 3.61, averaging 1.14) (Fig. 7b).

Table 2

Major (%) and trace element ($\times 10^{-6}$) concentrations of garnet at the Wanlongshan Zn-Sn polymetallic deposit by LA-ICP-MS analysis

Sample No.	15 [#] -W(Grt)													
	1	2	3	4	5	6	7	8	9	10	11	12	13	14
SiO ₂	37.04	36.79	36.39	36.79	36.31	37.14	35.96	37.16	35.94	36.19	36.22	36.62	36.67	35.63
TiO ₂	0.22	0.26	0.36	0.20	0.20	0.15	0.13	0.12	0.13	0.14	0.12	0.07	0.08	0.13
Al ₂ O ₃	16.93	17.12	16.92	16.65	16.98	15.95	11.58	16.17	11.09	16.43	16.89	17.32	16.16	12.89
FeO ^T	8.25	8.80	8.83	8.76	8.47	9.36	16.41	9.59	17.01	9.56	9.08	8.26	9.88	15.49

MnO	1.51	1.59	1.45	1.58	1.89	1.46	1.43	1.81	1.32	1.57	1.56	1.64	1.52	1.58
MgO	0.18	0.08	0.12	0.10	0.12	0.07	0.05	0.06	0.10	0.07	0.09	0.13	0.08	0.08
CaO	34.99	34.83	35.21	35.24	35.34	35.25	33.58	34.69	33.44	35.45	35.37	35.19	34.90	33.40
La	0.004	0.01	0.01	0.004	0.01	0.01	0.05	0.03	0.02	0.01	0.01	0.004	0.02	0.02
Ce	0.06	0.10	0.16	0.12	0.11	0.17	0.47	0.16	0.50	0.15	0.12	0.14	0.16	0.18
Pr	0.05	0.07	0.09	0.10	0.09	0.12	0.23	0.07	0.29	0.10	0.10	0.09	0.10	0.09
Nd	0.59	1.41	1.57	1.29	0.96	1.23	2.30	0.80	4.27	1.23	1.12	1.26	1.17	1.41
Sm	0.47	1.44	1.23	0.40	0.41	0.38	1.13	0.47	2.63	0.36	0.29	0.33	0.28	1.48
Eu	0.17	0.54	0.44	0.10	0.04	0.06	0.27	0.05	0.84	0.12	0.11	0.07	0.07	1.83
Gd	0.49	1.42	0.88	0.29	0.19	0.23	0.67	0.57	1.42	0.18	0.17	0.14	0.13	1.60
Tb	0.09	0.24	0.19	0.05	0.04	0.02	0.11	0.16	0.15	0.03	0.03	0.02	0.02	0.29
Dy	0.71	1.50	1.19	0.20	0.23	0.12	0.60	1.25	0.78	0.20	0.16	0.12	0.08	1.77
Ho	0.12	0.25	0.23	0.03	0.04	0.02	0.09	0.26	0.13	0.03	0.03	0.02	0.01	0.27
Er	0.39	0.72	0.68	0.11	0.10	0.08	0.28	0.79	0.34	0.08	0.10	0.06	0.05	0.75
Tm	0.05	0.08	0.09	0.01	0.01	0.00	0.04	0.09	0.05	0.01	0.01	0.01	0.01	0.10
Yb	0.44	0.58	0.60	0.09	0.11	0.06	0.17	0.57	0.35	0.05	0.05	0.03	0.03	0.82
Lu	0.04	0.06	0.08	0.01	0.01	0.01	0.02	0.06	0.04	0.00	0.01	0.00	0.00	0.09

Y	3.39	6.58	6.30	0.93	1.05	0.62	2.53	6.47	3.67	0.74	0.62	0.57	0.38	9.78
ΣREE	3.69	8.41	7.43	2.80	2.35	2.52	6.42	5.33	11.82	2.56	2.30	2.30	2.14	10.72
LREE	1.34	3.56	3.50	2.00	1.62	1.97	4.45	1.59	8.55	1.97	1.74	1.89	1.80	5.01
HREE	2.35	4.85	3.94	0.80	0.73	0.55	1.97	3.75	3.27	0.59	0.56	0.40	0.34	5.71
LREE/HREE	0.57	0.73	0.89	2.50	2.23	3.61	2.26	0.42	2.62	3.34	3.13	4.69	5.35	0.88
La _N /Yb _N	0.01	0.01	0.01	0.04	0.05	0.13	0.21	0.04	0.05	0.07	0.10	0.08	0.38	0.02
δEu	1.06	1.14	1.22	0.88	0.42	0.57	0.89	0.29	1.20	1.33	1.46	0.85	1.00	3.61
δCe	0.38	0.45	0.51	0.36	0.38	0.43	0.57	0.62	0.52	0.46	0.39	0.47	0.48	0.59
Rb	–	0.02	0.01	0.04	0.05	–	0.12	–	0.03	–	0.14	0.03	0.76	0.52
K	–	1.05	7.48	–	–	–	0.46	–	2.76	1.17	4.09	1.05	21.2	22.8
Ba	0.03	0.08	0.01	0.001	0.06	0.02	0.05	0.03	0.02	0.05	0.08	0.02	0.23	0.06
Th	0.22	0.14	0.30	0.06	0.10	0.14	0.15	0.11	0.04	0.07	0.10	0.14	0.12	0.02
U	2.83	2.83	4.49	1.69	0.87	1.25	0.66	0.08	0.24	1.49	1.52	1.59	1.39	0.52
P	18.2	29.7	49.7	44.9	13.8	22.9	31.6	39.4	28.8	17.5	52.1	28.2	34.9	39.1
Ta	1.33	0.93	0.91	1.16	2.35	0.77	1.02	4.97	2.83	0.57	0.65	0.40	0.80	2.95
Nb	34.5	46.5	61.5	21.6	23.9	19.5	11.6	7.02	11.3	17.5	13.7	9.11	9.70	17.4
Sr	0.01	0.12	0.04	0.02	0.05	0.04	0.07	0.03	0.004	0.02	0.64	0.03	0.34	0.12

Ti	1323	1561	2163	1175	1191	913	777	696	769	863	706	442	486	796
Zr	18.7	14.0	24.2	4.22	5.07	2.58	10.5	10.6	23.5	2.81	2.46	1.91	1.73	21.3
Hf	0.50	0.43	0.83	0.13	0.26	0.09	0.13	0.25	0.69	0.10	0.09	0.07	0.07	0.87
Zn	5.09	5.15	5.72	14.6	5.25	4.30	4.53	4.22	4.98	3.89	5.19	4.55	6.55	5.91
Mo	0.04	0.11	0.09	0.07	0.14	0.07	0.08	0.09	0.05	0.04	0.04	0.11	0.10	0.07
Sn	4788	2071	3303	3181	3466	2619	3013	946	3776	2381	3025	4067	3226	2447
Pb	0.01	–	0.07	0.002	0.16	0.04	0.24	0.03	0.11	0.01	0.31	0.004	1.28	1.78

Abbreviation: – = data unavailable.

5.3. Trace element composition of sphalerite

LA-ICP-MS analyses were conducted on two sphalerite samples across 21 analysis spots. The results are presented in Table 3 reveal iron content $w(\text{Fe})$ ranging from 8.30% to 15.18% (average 9.42%), with all values exceeding 8.0%, definitively classifying these samples as marmatite. The zinc content $w(\text{Zn})$ varies from 50.70% to 58.09% (mean 56.85%). The manganese content $w(\text{Mn})$ ranges from 396×10^{-6} to 2225×10^{-6} (average 963×10^{-6}), while copper content $w(\text{Cu})$ spans from 19.7×10^{-6} to 1957×10^{-6} (average 384×10^{-6}). The cadmium content $w(\text{Cd})$ varies between 762×10^{-6} and 1300×10^{-6} (average 1153×10^{-6}), and the indium content $w(\text{In})$ ranges from 652×10^{-6} to 1010×10^{-6} (average 929×10^{-6}). Additional elements (e.g., Co, Ni, Ga, Ge, Ag, Sn, Pb) are present in comparatively lower concentrations. The average concentrations are as follows: $w(\text{Co})$ at 4.94×10^{-6} ; $w(\text{Ni})$ at 0.59×10^{-6} ; $w(\text{Ga})$ at 1.04×10^{-6} ; $w(\text{Ge})$ at 1.71×10^{-6} ; $w(\text{Ag})$ at 3.37×10^{-6} ; $w(\text{Sn})$ at 4.48×10^{-6} ; and $w(\text{Pb})$ at 1.49×10^{-6} . Collectively, the trace elemental signature of the Wanlongshan sphalerite reflects enrichment in Fe, Mn, Cu, Cd, and In, with relative depletion in Ga, Ge, Sn, and Pb.

Table 3

LA-ICP-MS results of Sphalerite in the Wanlongshan Zn-Sn polymetallic deposit ($\times 10^{-6}$)

Sample No.	Mn	Fe	Co	Ni	Cu	Ga	Ge	Ag	Cd	In	Sn	Pb	Zn
52-1-Sp1	861	84834	4.59	1.21	1957	0.15	3.56	3.23	1188	633	1.10	1.57	576832
52-1-Sp2	1007	92139	4.66	0.45	65	0.32	–	2.01	1300	696	0.99	0.73	570778

52-1-Sp3	983	91334	4.87	0.10	36	0.00	1.65	1.58	1233	657	2.31	0.27	571767
52-1-Sp4	1039	95590	5.21	0.57	20	0.15	0.54	1.15	1215	657	0.61	0.52	567258
52-2-Sp1	896	88373	4.47	0.91	295	1.93	3.44	2.90	1176	605	0.69	0.15	574796
52-2-Sp3	753	83391	5.42	0.70	144	2.06	–	3.42	1165	622	1.03	0.96	580332
52-2-Sp4	992	93558	5.50	0.87	118	2.28	2.22	2.32	1242	649	0.58	1.58	569321
52-2-Sp5	917	89513	4.65	0.19	48	0.59	–	2.24	1117	648	1.82	1.89	573840
52-1-Sp1	970	90951	5.55	0.26	355	0.69	0.31	3.02	1102	632	1.64	2.00	571990
52-1-Sp3	672	85991	4.88	1.22	545	0.25	–	2.89	1046	567	6.87	3.08	577381
52-1-Sp4	984	92108	5.41	0.69	477	0.26	–	3.35	1149	630	0.92	2.35	570595
52-1-Sp6	965	91008	5.98	0.24	59	0.22	–	2.16	1132	630	0.28	0.50	572209
52-1-Sp7	686	82966	5.22	0.57	103	0.30	0.77	2.12	1115	606	1.14	2.09	580925
52-2-Sp1	1001	95926	4.55	0.04	64	0.93	1.21	2.46	1197	710	0.54	0.60	566859
52-2-Sp2	998	93165	4.20	0.59	23	1.46	–	1.82	1214	667	–	0.74	569838
52-2-Sp3	1016	92766	4.89	0.04	1173	1.85	–	2.82	1191	691	1.48	1.50	569075
52-2-Sp4	1010	93832	4.13	0.00	70	3.09	–	1.70	1173	672	0.88	1.07	569103
52-2-Sp5	904	88248	4.29	0.17	178	0.79	–	1.86	1178	656	0.76	0.14	575004
52-2-Sp6	952	89668	3.96	0.65	1107	2.44	0.81	3.57	1189	629	0.50	0.60	572529

Abbreviation: – = data unavailable.

5.4. *In situ* S isotopes

In this study, in-situ sulfur isotope analysis was conducted on various types of sulfides (pyrite, chalcopyrite, and sphalerite) from the main mineralization stage of the Wanlongshan Zn-Sn polymetallic deposit.

We analyzed a total of 38 points, including 11 on pyrite, 7 on chalcopyrite, and 20 on sphalerite (Table 4). The $\delta^{34}\text{S}$ values for pyrite demonstrate a relatively narrow distribution, ranging from 0‰ to 3.8‰ (mean=1.5 ‰, range=3.8 ‰). Chalcopyrite exhibits a more constrained range of 2.0‰ to 2.5‰ (mean=2.2 ‰, range=0.5 ‰). Sphalerite shows a $\delta^{34}\text{S}$ range from 1.9‰ to 2.9‰ (mean=2.2 ‰, range=0.5 ‰).

Table 4

$\delta^{34}\text{S}$ isotope analysis results of the Wanlongshan Zn-Sn polymetallic deposit

Sample No.	Mineral	$\delta^{34}\text{S}_{\text{V-CDT}}\text{‰}$	Sample No.	Mineral	$\delta^{34}\text{S}_{\text{V-CDT}}\text{‰}$
52-1-Ccp-1	chalcopyrite	2.1	52-2-Sp-6	sphalerite	2.5
52-1-Ccp-2	chalcopyrite	2.5	52-2-Sp-7	sphalerite	2.4
52-1-Ccp-3	chalcopyrite	2.2	2-Sp-1	sphalerite	2.2
52-1-Ccp-4	chalcopyrite	2.2	2-Sp-2	sphalerite	2.4
52-1-Ccp-5	chalcopyrite	2.0	2-Sp-3	sphalerite	2.5
52-1-Ccp-6	chalcopyrite	2.3	2-Sp-4	sphalerite	2.0
52-1-Ccp-7	chalcopyrite	2.4	2-Sp-5	sphalerite	2.7
52-1-Sp-1	sphalerite	2.3	2-Sp-6	sphalerite	2.5
52-1-Sp-2	sphalerite	2.4	2-1-Py-1	Pyrite	3.8
52-1-Sp-3	sphalerite	2.5	2-1-Py-2	Pyrite	3.4
52-1-Sp-4	sphalerite	2.6	2-1-Py-3	Pyrite	2.5

52-1-Sp-5	sphalerite	2.2	2-1-Py-4	Pyrite	2.2
52-1-Sp-6	sphalerite	2.6	2-1-Py-5	Pyrite	2.0
52-1-Sp-7	sphalerite	2.4	2-2-Py-1	Pyrite	0.6
52-2-Sp-1	sphalerite	2.2	2-2-Py-2	Pyrite	0.0
52-2-Sp-2	sphalerite	2.9	2-2-Py-3	Pyrite	0.9
52-2-Sp-3	sphalerite	2.3	2-2-Py-4	Pyrite	0.7
52-2-Sp-4	sphalerite	1.9	2-2-Py-5	Pyrite	0.4
52-2-Sp-5	sphalerite	2.0	2-2-Py-6	Pyrite	0.5

6. Discussion

6.1. Trace element composition of garnets and its significance

6.1.1. Compositional characteristics of REEs in garnet and influencing factors

Research has shown that REE partitioning in garnet is influenced by multiple factors, including material provenance, fluid physicochemical attributes, crystallographic structure, and electrostatic intramineral forces. REEs incorporated into garnet predominantly through four mechanisms: surface adsorption, absorption, isomorphous substitution, and interstitial occupancy within solid solutions (Smith et al., 2004). The standard crystallographic formula for garnet, $X_3Y_2Z_3O_{12}$, features octahedrally coordinated divalent cations (Ca^{2+} , Mg^{2+} , Mn^{2+} , Fe^{2+}) in the X position, with incorporation governed by ionic radii and charge. REEs primarily enter garnet by substituting for X^{2+} cations in octahedral sites (Gaspar et al., 2008; Yang et al., 2019).

The chemical composition of garnet significantly influences its REE distribution patterns. Typically, grossular-rich garnets show HREE enrichment and LREE depletion, while andradites display relative LREE enrichment (Gaspar et al., 2008). The garnets from the Wanlongshan Zn-Sn polymetallic deposit display a left-skewed REE distribution pattern (Fig. 7), potentially governed by their grossular-rich composition. However, a conspicuous linear correlation between REEs and the concentrations of Ca, Mg, and Fe is absent (Fig. 8a–c), suggesting that the partitioning of REEs within the garnet may not be solely dictated by the crystallographic structure of the garnet but may also be subject to the influence of fluid composition and intragranular electrostatic interactions.

6.1.2. Formation conditions of garnet

(1) Oxygen fugacity

In the initial phase of skarn genesis, contact metamorphic processes occur between intrusive magma and surrounding rocks, resulting in the formation of anhydrous silicate minerals such as grossular, andradite, and diopside). Experimental investigations have

demonstrated that grossular typically forms in fluids that are weakly oxidizing to weakly reducing, whereas andradite, which requires a higher concentration of Fe^{3+} to fill the Y^{3+} sites within the $(\text{Y}^{3+}(\text{X}_3\text{Y}_2[\text{SiO}_4]_3))$ framework, is more prone to crystallize in solutions that are oxidizing to weakly oxidizing (Ai and Jin, 1981; Zhao et al., 1983). Consequently, the Fe^{3+} concentration serves as an indicator of the oxygen fugacity of the fluid. Observations from Table 2 indicate that the garnets from the Wanlongshan Zn-Sn polymetallic deposit are relatively iron-deficient. They exhibit an average Fe^{3+} content of only 0.63%. This low iron content implies formation under conditions of low oxygen fugacity.

The trace element composition of garnet can be indicative of the oxygen fugacity during its formation. In oxidizing conditions, Eu predominantly exists as Eu^{3+} , but under more reducing conditions, Eu^{2+} , with an ionic radius akin to that of Ca^{2+} , is more likely to substitute for Ca^{2+} within the garnet structure. The chondrite-normalized pattern of REEs for the garnets from the Wanlongshan Zn-Sn polymetallic deposit (Fig. 6) reveals an avg. δEu of 1.14, which overall suggests a weak anomaly, collectively pointing to the formation of these garnets under low oxygen fugacity conditions within a weakly oxidizing to weakly reducing environment.

(2) pH conditions

Grossulars garnets typically form in moderately intermediate-acidic fluids at temperatures of 550 to 700°C, while andradites crystallize in moderately alkaline solutions ranging from 450 to 600°C (Ai and Jin, 1981). Fluid activity significantly influences REE fractionation depending on pH conditions. Under near-neutral pH, garnet REE patterns characteristically show LREE depletion and HREE enrichment, accompanied by either negative or inconspicuous Eu anomalies. In weakly acidic conditions, Cl^- ions substantially impact REE patterns (Bau, 1991). The formation of the stable complex EuCl_4^{2-} through the combination of Cl^- with Eu^{2+} enhances the stability of Eu^{2+} in the fluid, increasing its concentration and leading to a positive europium anomaly in garnet (Mayanovic et al., 2007; Gaspar et al., 2008). Garnets from the Wanlongshan Zn-Sn polymetallic deposit exhibit pronounced differentiation between LREE and HREE, with LREE/HREE ratios varying from 0.42 to 5.35 (Fig. 8d), and exhibit weak europium anomalies. Based on these characteristics, it is inferred that these garnets likely formed under neutral to weakly acidic conditions.

(3) Metasomatism

Skarns are predominantly generated through two metasomatic processes: diffusion and infiltration. Previous investigations have established that under closed-system conditions characterized by low water/rock ratios and acidic to weakly acidic fluids, the dominant metasomatic process is diffusion, which correlates with slower rates of mineral crystallization. Conversely, under open-system conditions with higher water/rock ratios—where fluids are primarily transported as chloride complexes (Bau, 1991; Smith et al., 2004; Gaspar et al., 2008)—the primary metasomatic process is infiltration. This process accelerates the rate of mineral crystallization and can lead to the formation of pronounced oscillatory zoning in garnets. The garnets from the Wanlongshan Zn-Sn polymetallic deposit lack such oscillatory zoning, and the grossular exhibits anisotropy, as evidenced by abnormal interference colors or twinning phenomena under orthogonal polarized light. The absence of numerous fluid inclusions within the minerals implies a slower growth rate and a low water/rock ratio (Fig. 5e), suggesting their formation in a relatively stable and closed system where diffusional metasomatism predominates. Considering the spatial distribution of the skarn and the granite, it is inferred that the magmatic-hydrothermal fluid and the schist interact through diffusional metasomatism in a relatively sealed environment, resulting in the formation of garnets devoid of zoning.

6.2. Trace element composition of sphalerite and its implications for ore formation conditions

6.2.1. Implications of trace element composition for ore-forming temperatures

Investigations have demonstrated substantial differences in the trace element content of sphalerite crystallized at varying temperatures. Sphalerite formed under elevated temperature conditions is typically enriched in elements such as Fe, Mn, and In, whereas sphalerite precipitated under lower temperature conditions shows an enrichment in elements such as Cd, Ga, and Ge (Kelley et al., 2004; Frenzel et al., 2016). Within the Wanlongshan deposit, sphalerite exhibits a relative enrichment in Fe, Mn, and In, with an avg. concentrations of $94,218 \times 10^{-6}$, 963×10^{-6} , and 629×10^{-6} , respectively, and is depleted in Ga (average = 1.04×10^{-6}) and Ge (average = 1.71×10^{-6}) (Table 3). These features indicate that the sphalerite in the deposit originated under medium to high-temperature conditions, aligning with the geological characteristics of the deposit.

The concentration of Fe in sphalerite is typically positively correlated with the formation temperature, with a higher Fe/Zn ratio suggesting a higher temperature of ore genesis (Keith et al., 2014; Frenzel et al., 2016). Fe/Zn ratios in Wanlongshan sphalerite range from 0.14 to 0.30, with an average of 0.16. The Manjiazhai main ore-forming period shows higher values, between 0.18 and 0.28 (averaging 0.23). In contrast, Nandangchang sphalerite exhibits much lower ratios from 0.02 to 0.11, averaging only 0.06 (unpublished data). This suggests a decreasing trend in ore-forming temperature from Manjiazhai to Wanlongshan to Nandangchang, along with an increasing distance from the Laojunshan granite, aligning with the pattern that closer proximity to the magmatic body correlates with higher ore-forming temperatures. Utilizing the empirical thermometer formula for sphalerite, $\text{Fe/Zn (sphalerite)} = 0.0013t - 0.2953$ (Keith et al., 2014), the minimum fluid precipitation temperature for Wanlongshan sphalerite is estimated to be between 337 and 457°C, averaging 355°C. The Zn/Cd and Ga/In ratios within sphalerite can indicate the ore-forming temperature, where high-temperature environments are characterized by Zn/Cd ratios greater than 600 and Ga/In values generally less than 0.01; medium-temperature environments have Zn/Cd ratios between 100 and 600 and Ga/In values ranging from 0.01 to 5; and low-temperature environments exhibit Zn/Cd ratios less than 100 and Ga/In values between 1 and 100 (Kang et al., 2020). In the Wanlongshan deposit, the Zn/Cd and Ga/In values of sphalerites range from 463 to 665 and 0.0002 to 0.014, with mean values of 496 and 0.002, respectively, both indicating formation in a medium to high-temperature setting.

6.2.2. Replacement mechanisms of trace element in sphalerite

A plethora of studies have discussed that the replacement of Zn by trace elements in sphalerite is predominantly facilitated through single-element or multi-element coupled substitution mechanisms. In the single-element substitution mechanism, several elements within sphalerite possess ionic radii analogous to that of Zn, enabling direct substitution of Zn^{2+} in the sphalerite crystal lattice, such as Fe^{2+} , Cu^{2+} , Mn^{2+} , and Cd^{2+} (Cook et al., 2009). For the multi-element coupled substitution mechanism, the combined valences of multiple elements equate to the combined valences of Zn, exemplified by $\text{Ge}^{4+} + 2(\text{Cu}^+, \text{Ag}^+)$ substituting for 3Zn^{2+} (Wu et al., 2019), $(\text{Ga}^{3+} + \text{Cu}^+)$ replacing 2Zn^{2+} (Wang et al., 2021), and the combination of $(\text{Cu}^+, \text{Ag}^+) + (\text{Ga}^{3+}, \text{As}^{3+})$ substituting for 2Zn^{2+} (Yang et al., 2021). Trace elements in sphalerite may also be present as minute inclusions, which manifest as anomalous peaks in the time-resolved profile of LA-ICP-MS analyses. In contrast, a smooth curve signal suggests presence through isomorphous substitution (Ye et al., 2011). Correlation diagrams of the trace elements within sphalerite can more effectively illustrate the inter-element substitution relationships (Fig. 9).

In the time-resolved profiling of sphalerite, Fe, Mn, and Cd display smooth signals that are comparable to those of Zn and S (Fig. 10). Linear correlation exist between Fe, Mn, and Cd (Fig. 9a, b), with a strong positive correlation between Zn and the combined concentrations of Mn, Cd, and Fe ($R^2=0.98$) (Fig. 9c), suggesting that Fe^{2+} , Mn^{2+} , and Cd^{2+} can directly substitute for Zn^{2+} within the sphalerite lattice. Due to the considerable difference in ionic radii between Pb and Zn, Pb is commonly found as minute inclusions within sphalerite. The smooth signal of Pb in the time-resolved profile of sphalerite exhibits a peak (Fig. 10a), and the concentration of Pb shows significant variation (Table 3),

indicating that Pb likely occurs in sphalerite as minute inclusions, such as galena or other lead-bearing minerals. Additionally, the substantial fluctuation range of Cu in the time-resolved profile (Fig. 10b) indicates considerable Cu incorporation into the sphalerite lattice, corroborated by the frequent presence of "chalcopyrite disease" within the crystal (Fig. 5f). Research has demonstrated that the solubility of Cu in sphalerite is relatively low, preventing the formation of a significant amount of chalcopyrite solid solution through exsolution, and the introduction of external hydrothermal fluids may lead to the development of "chalcopyrite disease" structures (Kojima and Sugaki, 1985). Experimental studies have further demonstrated that "chalcopyrite disease" predominantly occurs in iron-rich sphalerite, with excess sulfur and iron, where the fluid serves as the Cu source (Govindarao et al., 2018).

Cations with oxidation states of +1, +3, and +4 typically incorporate into the sphalerite lattice through complex substitution mechanisms replacing Zn^{2+} ions (Zhuang et al., 2019). Ag displays a notable positive correlation with Cu ($R^2=0.38$) (Fig. 9d), indicating that Ag may replace a portion of Cu, with substitution mechanisms such as $(\text{Ag}^+ + \text{Ga}^{3+})$ replacing 2Zn^{2+} and $(2\text{Ag}^+ + \text{Ge}^{4+})$ replacing 3Zn^{2+} . Under reducing conditions, Ga with a coordination number of six has an ionic radius that is similar to that of elements (e.g., Zn, Sn, Cu, Fe, and Sb) in sulfide deposits, particularly akin to Zn, leading to sphalerite in such deposits being a major host for Ga enrichment. Cook et al. (2009) reported a positive correlation between Ga and the combined concentrations of (Cu + Ag), suggesting that Ga predominantly enters sphalerite in conjunction with Cu and Ag, with the substitution mechanism being $2\text{Zn}^{2+} \leftrightarrow \text{Ga}^{3+} + (\text{Ag}, \text{Cu})^+$. Li and Zhu (2020) discovered that in the Huize Pb-Zn deposit, the Ga content in sphalerite closely matches that of In, inferring that Ga may also enter sphalerite in conjunction with In to substitute for Zn, represented as $3\text{Zn}^{2+} \leftrightarrow \text{Ga}^{3+} + \text{In}^{3+}$. Ga in the Wanlongshan sphalerite is not enriched, with an avg. concentration of merely 1.04×10^{-6} , and there is no significant correlation with Zn, Cu, Ag, and In (Fig. 9e-h), indicating that only a minor amount of Ga is incorporated into the sphalerite.

Research has identified potential substitution mechanisms for In within sphalerite, which are as follows: ① $\text{Cu}^+ + \text{In}^{3+} \leftrightarrow \text{Zn}^{2+} + \text{Fe}^{2+}$ (Johan, 1988); ② $(\text{Cu}, \text{Ag})^+ + \text{In}^{3+} \leftrightarrow 2\text{Zn}^{2+}$ (Cook et al., 2009; Ye et al., 2011, Murakami and Ishihara, 2013); ③ $\text{In}^{3+} + \text{Sn}^{3+} + (\bullet) \leftrightarrow 3\text{Zn}^{2+}$ (Belissont et al., 2014); ④ $\text{In}^{3+} + (\text{Cu}, \text{Ag})^+ + \text{Sn}^{2+} \leftrightarrow 3\text{Zn}^{2+}$ or $\text{In}^{3+} + \text{Sn}^{4+} + (\text{Cu}, \text{Ag})^+ + (\bullet) \leftrightarrow 4\text{Zn}^{2+}$ (Frenzel, et al., 2016), where $\bullet$ represents a vacancy. In the Wanlongshan sphalerite, Cu and In concentrations are strongly correlated, with the majority of the plotted data points aligning with or in proximity to the 1:1 Cu-In ratio line (Fig. 9i). This correlation suggests a co-enrichment of Cu and In within the sphalerite, with the dominant substitution mechanism likely being $\text{Cu}^+ + \text{In}^{3+} \leftrightarrow 2\text{Zn}^{2+}$.

6.2.3. Implications of mineral deposit genetic types from trace element compositions of sphalerite

The origin of the Dulong Sn-Zn polymetallic deposit within the Laojunshan ore cluster in southeastern Yunnan Province remains a contentious issue, with competing theories including sedimentary exhalative, magmatic-hydrothermal, and multiphase mineralization (Zhou et al., 1998; Jia et al., 2014; Liu et al., 1998, 2007). Recent research has increasingly pointed to a strong correlation between the Dulong Sn-Zn polymetallic deposit and the late Yanshanian granite. The magmatic-hydrothermal activities and their interplay with the country rocks are likely to have a determinative influence on the mineralization process (Yang et al., 2019).

Fig. 11 provides a comparative analysis of the trace element composition in sphalerite from diverse Pb-Zn ore deposits. Mississippi Valley-type (MVT) Pb-Zn deposits throughout the Sichuan-Yunnan-Guizhou region characteristically contain sphalerite enriched in Ga, Ge, and Cd while depleted in Fe, Mn, and In (Ye et al., 2011). In contrast, Sedex deposits (e.g., Laochang, Yunnan, and Dabaoshan, Guangdong) exhibit sphalerite enriched in Fe, Sn, Mn, and Cd, yet impoverished in Co and Ni (Ye et al., 2012). The

sphalerite of skarn deposits (e.g., Gejiu, Hetaoping, and Huanggangliang) are enriched in Fe, Mn, and In, and depleted in Cd, Ge, and Ga (Ye et al., 2011; Li et al., 2015). Meanwhile, the VHMS deposits (e.g., Eskay Creek in Canada), are distinguished by sphalerite that is enriched in Fe, Mn, and Cd, but impoverished in In and Sn (Cook et al., 2009). Despite the lower concentrations of Ge and Ga in sphalerite from the Wanlongshan deposit relative to these classic examples (Fig. 11), its relative enrichment in Fe, Mn, and Cd aligns with that of skarn deposits (e.g., Gejiu and Hetaoping), implying a predominant hydrothermal metasomatic origin for the deposit.

In the plots of Ga-Fe, Ga-Mn, Fe/Mn-Mn/Ga, and Fe/Mn-Fe/100Ga relationships for different sphalerite types (Fig. 12). All data points plot firmly within the established field for skarn deposits. This pattern clearly distinguishes the Wanlongshan mineralization from other deposit types such as Sedex, MVT, and VHMS, which would show distinctly different geochemical signatures in these diagrams.

6.3. Sources of Sulfur isotope

The study of sulfur isotope composition is crucial for determining the source of ore-forming materials (Ohmoto, 1986). Sulfur sources in sulfide deposits are typically categorized into three types (Ohmoto, 1972, 1986): (1) mantle-derived sulfur with $\delta^{34}\text{S}$ value near zero, exhibiting a tower-like distribution with a range of $0 \pm 3\text{‰}$; (2) crustal-derived sulfur, with a $\delta^{34}\text{S}$ value around 20‰ , characterized by a wide range of variation, with marine or marine sulfates enriched in heavy sulfur and biogenic sources enriched in light sulfur, and (3) mixed sulfur, exhibiting a combined $\delta^{34}\text{S}$ signature. Discussions on the S source in deposits must be based on the $\delta^{34}\text{S}_{\text{SS}}$ values. However, obtaining an accurate value of $\delta^{34}\text{S}_{\text{SS}}$ in the ore-forming hydrothermal fluid is challenging in hydrothermal deposits, as the S isotope composition of hydrothermal minerals depends not only on the $\delta^{34}\text{S}$ value of the source material but also on the physical and chemical conditions during the migration of sulfur-bearing materials and mineral precipitation (Ohmoto, 1972). Due to the isotope fractionation effect between solid and liquid phases in hydrothermal ore-forming processes, the $\delta^{34}\text{S}$ values of sulfides and sulfates formed by hydrothermal fluids are generally not equal to the $\delta^{34}\text{S}_{\text{SS}}$ values of the hydrothermal fluid, but rather a function of the $\delta^{34}\text{S}_{\text{SS}}$, f_{O_2} , pH, ionic strength, and temperature ($\delta^{34}\text{S} = f(\delta^{34}\text{S}_{\text{SS}}, f_{\text{O}_2}, \text{pH}, I, T)$), and only in cases with simple mineral assemblages and an absence of sulfate minerals can the $\delta^{34}\text{S}$ value of sulfides roughly represent the $\delta^{34}\text{S}_{\text{SS}}$ values of the hydrothermal fluid (Ohmoto and Rye, 1979). The Wanlongshan Zn-Sn polymetallic deposit features a simple sulfide assemblage, primarily chalcopyrite, pyrite, and sphalerite, without sulfate minerals like barite. This indicates low oxygen fugacity in the ore-forming fluid, with sulfur mainly existing as divalent H_2S . The consistent $\delta^{34}\text{S}$ values across different sulfide minerals suggest minimal sulfur isotope fractionation during hydrothermal activity, allowing the $\delta^{34}\text{S}$ of the sulfides to represent the $\delta^{34}\text{S}_{\text{SS}}$ of the ore-forming hydrothermal fluid.

In general, the $\delta^{34}\text{S}$ values of various minerals analyzed in this study range from 0 to 3.8‰ (mean 2.1‰ , $n=38$), markedly lower than the S isotopic composition of Cambrian seawater (15‰ to 35‰ , Claypool et al., 1980) and the $\delta^{34}\text{S}$ values of pyrite from the Cambrian strata in the region (17.5‰ to 25.8‰ , He et al., 2014). This precludes the host strata from contributing significant sulfur. The $\delta^{34}\text{S}$ values are relatively consistent, exhibiting a pronounced "tower effect" akin to the isotopic composition of magmatic sulfur. Some pyrite samples with slightly elevated $\delta^{34}\text{S}$ values (3.8‰ and 3.4‰) suggest possible minor incorporation of additional sulfur sources, potentially related to stratigraphic contamination, though not substantial. Overall, the evidence indicates that sulfur in the ore-forming fluid predominantly derived from magmatic-hydrothermal sources.

6.4. Skarn formation age and ore deposit genesis

The Wanlongshan Zn-Sn polymetallic deposit, a significant deposit within the Dulong ore field, plays a crucial role in understanding rock-forming and metallogenic mechanisms (Lin et al., 2016).

Precisely determining the ore-forming timeframe is essential for elucidating the relationship between the mineralization epochs and regional geological evolution, which is vital for understanding ore-forming regularities and predicting mineralization zones. Traditional geochronological methods for skarn deposits typically involve U-Pb dating of zircons from intrusive bodies to ascertain rock formation ages and Re-Os dating of molybdenite to pinpoint ore-forming ages. However, accurate geochronological techniques specifically for skarn formation have been lacking.

Garnet U-Pb dating results reveal the formation at 88.1 ± 3.3 Ma (MSWD=0.98, $n=17$) (Fig. 6), representing the crystallization age of the skarn in the Wanlongshan Zn-Sn polymetallic deposit. This age is consistent, within the error range, with the Re-Os dating of molybdenite from the mining district ($\sim 88.9 \pm 3.5$ Ma, Wang and Ai, 2018), the Sn-Zn mineralization ages in the northern Manjiazhai area (75~121 Ma, Liu et al., 1999, 2007; Li et al., 2013; Wang et al., 2014; Zhao, 2017; Liu, 2021), and the ages of the Laojunshan granite (80~97 Ma, Liu et al., 2007; Zhang et al., 2012; Li et al., 2013; Ma, 2013; Feng et al., 2013; Liu et al., 2014; Xu et al., 2015; Lan et al., 2016; Xue, 2019; Liu, 2021) (Fig. 14, Table 5). The simultaneous occurrence of the granite and Zn-Sn mineralization suggests a possible genetic connection, implying that the skarn and Zn-Sn mineralization may have originated from magmatic-hydrothermal metasomatism. Conversely, if the skarn in the Wanlongshan deposit were of sedimentary exhalative origin, its formation age would be nearly synchronous with the host Cambrian strata. Geochronological studies support that the Wanlongshan deposit is a magmatic-hydrothermal metasomatic skarn deposit.

The trace element signature of sphalerite from the Wanlongshan Zn-Sn polymetallic deposit aligns with characteristics typical of established skarn deposits, further confirming its classification as a skarn-type deposit. Additionally, sulfur isotope analysis indicates that the sulfur in the ore-forming fluid originated from magmatic-hydrothermal fluids. Combining the chronology of the garnet, the genetic discrimination results of sphalerite, and in-situ S isotope results, it is uniformly suggested that the Wanlongshan Zn-Sn polymetallic deposit is a typical skarn deposit.

Table 5

The geochronological date of the Zn-Sn polymetallic deposits and associated granites in the Dulong ore field

Category	Location	Rock type	Testing method	Age (Ma)	Data source
Mineralization	Manjiazhai	Sphalerite	Rb-Sr	79.08 ± 9.11	Liu et al., 1999
	Manjiazhai	Cassiterite	TIMS U-Pb	79.8 ± 3.2 , 82.0 ± 9.6	Liu et al., 2007
	Manjiazhai	Quartz vein greisen	Re-Os	75.04 ± 1.78 , 79.16 ± 2.43	Li et al., 2013
		Phlogopite	Ar-Ar	93.6 ± 1.1	
	Manjiazhai	Skarn ore	LA-ICP-MS U-Pb	88.0 ± 1.6 , 89.2 ± 4.1	Wang et al., 2014
		Skarn ore	LA-ICP-MS U-Pb	87.2 ± 3.9	

	Tongjie - Manjiazhai	Cassiterite	LA-ICP-MS U-Pb	88±5	Zhao, 2017
	Manjiazhai	Cassiterite	LA-ICP-MS U-Pb	89.4±1.4	
	Wanlongshan	Molybdenite	Re-Os	88.9±3.5	Wang and Ai, 2018
	Manjiazhai	Cassiterite	LA-ICP-MS U-Pb	82.1±6.3~96.5±8.3	Liu, 2021
		Biotite	Ar-Ar	121.2±0.9	
		Muscovite	Ar-Ar	86.3±0.5	
	Wanlongshan	Garnet	LA-ICP-MS U-Pb	88.1±3.3	This Article
	Manjiazhai	Granite	Zircon U-Pb	82.0±9.6	Liu et al., 2007
		Blind granite	Zircon U-Pb	92.9±1.9	
		Granite porphyry	Zircon U-Pb	86.9±1.4	
Magmatism	Laojunshan	Monzogranite	Zircon U-Pb	96±2	Zhang et al., 2012
	Manjiazhai	Muscovite granite	Zircon U-Pb	87.6±1.4、91.7±1.5	Li et al., 2013
		Granite porphyry	Zircon U-Pb	87.3±2.1	
		Two-mica granite	Zircon U-Pb	84.3±2.2、85.0±1.3	
	Laojunshan	Two-mica monzogranite	Zircon U-Pb	85.88±0.76、88.0±2.0	Ma, 2013
	Manjiazhai	Granite	Zircon U-Pb	86.02±0.5~86.66±0.4	Feng et al., 2013
	Laojunshan	Two-mica granite	Zircon U-Pb	93.6±2.3	Liu et al., 2014
		Two-mica monzogranite	Zircon U-Pb	90.03±0.67、90.9±1.6	

		Biotite granite	Zircon U-Pb	89.24±0.55、93.1±1.6	
	Manjiazhai	Coarse-grained granite	Zircon U-Pb	90.1±0.7	Xu et al., 2015
		Fine-grained granite	Zircon U-Pb	89.7±0.8	
		Granite porphyry	Zircon U-Pb	86.0±0.5	
	Huashitou-Tongjie	Granite	Zircon U-Pb	88.3±1.1	
	Jinshipo	Granite porphyry	Zircon U-Pb	86.5±1.0	Lan et al., 2016
		Muscovite granite	Zircon U-Pb	97.3±1.1	
		Two-mica granite	Zircon U-Pb	95.9±0.9~117.1±0.9	
	Manjiazhai	Granite	Zircon U-Pb	82.8±1.6~87.5±0.6	Zhao, 2017
	Tongjie	Two-mica monzogranite	Zircon U-Pb	88.7±2.0、90.5±1.3	Xue, 2019
	Tongjie	Granite porphyry	Zircon U-Pb	88.4±2.9	
	Manjiazhai	Granite porphyry	Zircon U-Pb	83.9±1.1、84.48±0.43	

7. Conclusions

(1) Garnet from the Wanlongshan Zn-Sn polymetallic deposit displays characteristic LREE depletion and HREE enrichment, likely governed by crystal chemistry and adsorption effects. The composition of garnet primarily composed of grossular with a subdued Eu anomaly, indicates formation under conditions of low oxygen fugacity within a weakly oxidizing to weakly reducing, and neutral to slightly acidic environment, where hydrothermal metasomatism was predominantly driven by diffusion at low water/rock ratios.

(2) Sphalerite in the deposit is enriched in Fe, Mn, and In, and depleted in Ga, Ge, and Sn, suggesting formation at medium to high temperatures. Fe, Mn, and Cd substitute Zn through simple isomorphic replacement; Pb may be present as minute inclusions, and Cu is incorporated through chalcopyrite solid solution. The Ag substitution mechanism could involve Ag^+ paired with Ga^{3+} replacing 2Zn^{2+} or 2Ag^+ paired with Ge^{4+} replacing 3Zn^{2+} . Minimal Ga incorporation occurs, and the substitution mechanism for In is likely dominated by $\text{Cu}^+ + \text{In}^{3+} \leftrightarrow 2\text{Zn}^{2+}$.

(3) The U-Pb dating of garnet from the Wanlongshan Zn-Sn polymetallic deposit yields a formation age of 88.1 ± 3.3 Ma, correlating well with Re-Os molybdenite ages from the mining district, Sn-Zn mineralization ages in the northern Manjiazhai mining district, and emplacement ages of the Laojunshan granite within analytical error margins. This

temporal coincidence strongly suggests contemporaneous skarnization, Zn-Sn mineralization, and granite emplacement. The trace element compositions of both garnet and sphalerite, closely resembling geochemical signatures of established global skarn deposits, together with in-situ sulfur isotope data, conclusively establish the Wanlongshan Zn-Sn polymetallic deposit as a classic skarn-type mineralization system.

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Fig. 1. (a) Simplified tectonic map of China and (b) simplified geological map of southeastern Yunnan Province (modified after Mao et al., 2013).

Fig. 2. (a) Distribution of major tectonic units and geological map of the Laojunshan orefield in Yunnan Province. (b) Geological map of the Wanlongshan Zn-Sn polymetallic deposit. (modified after Xu et al., 2015)

Fig.3. Geological section along the exploration line 20 of the Wanlongshan Zn-Sn polymetallic deposit

Fig.4. Mineral paragenetic sequence of the Wanlongshan Zn-Sn polymetallic deposit

Fig. 5. The characteristics of the ore-fabric in the Wanlongshan Zn-Sn polymetallic deposit: (a) The interface between Marble (Mb), Skarn (Sk), and the ore body; (b) Symbiosis of Pyrite (Py), Sphalerite (Sp), and Chalcopyrite (Ccp); (c) Molybdenite (Mo) developed interstitially to Garnet (Grt) grains; (d) Symbiosis of Pyrite (Py), Sphalerite (Sp), and Pyrrhotite (Po); (e) Microscopic image of Garnet (Grt) captured under transmitted light; (f) Pyrite (Py), Chalcopyrite (Ccp), and Sphalerite (Sp) during the sulfide stage under reflected light; (g) Pyrite (Py), and Sphalerite (Sp) in the ZK2014 drilling; (h) Paragenesis of Pyrite(Py) and Sphalerite(Sp) in the orebody; (i) Boundary-intergrowth textures between Pyrite(Py) and Sphalerite(Sp). Abbreviations: Mb = Marble, Sk = Skarn, Grt = Garnet, Sp = Sphalerite, Ccp = Chalcopyrite, Py = Pyrite, Mo = Molybdenite, Po = Pyrrhotite

Fig.6. LA-ICP-MS U-Pb analytical spectra (a) and dating results (b) of garnet from the Wanlongshan Zn-Sn polymetallic deposit

Fig.7. Primitive mantle-normalized trace element concentrations (a) and chondrite normalized REE distribution patterns (b) of the Wanlongshan Zn-Sn polymetallic deposit. Primitive mantle-normalized values and chondrite-normalized values are from Sun and McDonough (1989). Additionally, trace elemental (including REE) compositions of the lower continental crust, upper continental crust and continental crust are also from Sun and McDonough (1989).

Fig.8. The relationship between trace elements in garnet at the Wanlongshan Zn-Sn polymetallic deposit (R refers to the correlation coefficient)

Fig.9. Correlation diagram of trace elements of sphalerite in the Wanlongshan Zn-Sn polymetallic deposit

Fig.10. Temporal resolution profiles of sphalerite at measuring points 8 and 15 in No. WL5 orebody

Fig. 11. Comparison of trace element contents of Sphalerite in the Wanlongshan Zn-Sn polymetallic deposit with different types of deposits MVT Pb-Zn deposits data are derived from Cook et al. (2009), Ye et al. (2011), and Hu et al. (2019). Sedex deposits from Ye et al. (2011). VHMS deposits from Cook et al. (2009) and Zhang et al. (2018). skarn deposits from Ye et al. (2011), Pi et al. (2015), Xu et al. (2017), and Li et al. (2015).

Fig.12. Binary plots of Ga vs. Fe (a), Ga vs. Mn (b), Fe/Mn vs. Mn/Ga (c) and Fe/Mn vs. Fe/100Ga (d) in different types of deposits(The data source is the same as Fig.11)

Fig. 13. (a) In-situ sulfur isotope distribution histogram of the Wanlongshan Zn-Sn polymetallic deposit and (b) sulfur isotope composition diagram of typical deposits in the Laojunshan ore district

Fig. 14. Age spectrum of diagenesis and mineralization in the Dulong ore field(date from Table 5)

- (1) The garnet is formed in neutral, or weakly acidic environments with low oxygen fugacity
- (2) The formation age of garnet is 88.1 ± 3.3 Ma.
- (3) Sphalerite from Wanlongshan is rich in Fe, Mn, and In, but poor in Ga, Ge, and Sn.

